



MATSim for Jakarta MRT Expansion: Projected Ridership and Mode Shifts

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Objective



To evaluate how MRT expansion reshapes travel behaviour and mobility outcomes through improved objective accessibility using an agent-based simulation framework in Jakarta TOD systems.

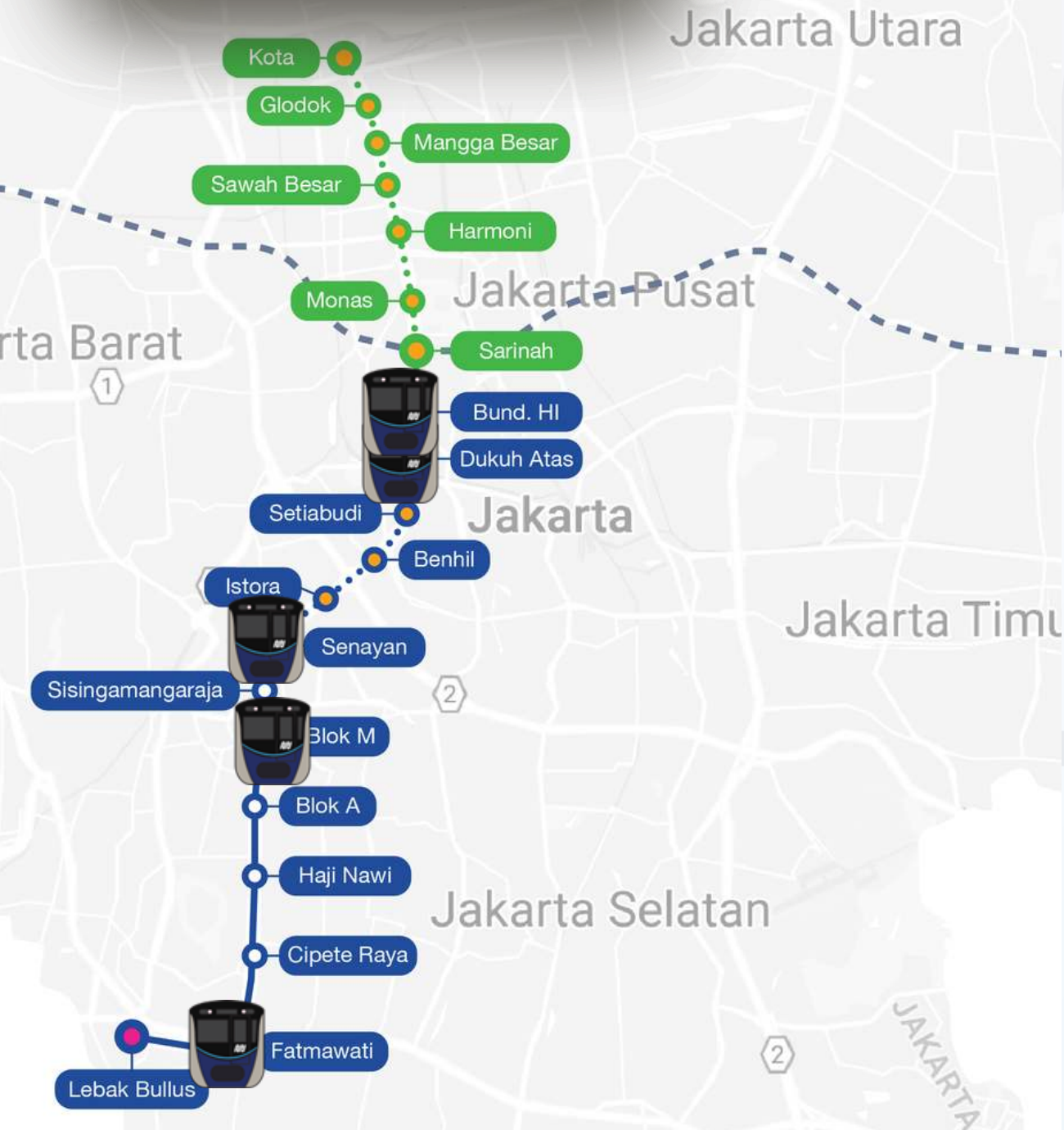
 Study Area

**MAP OF
INDONESIA**



Jakarta

(13+7) Station MRT Jakarta



15.7-kilometer line: 6 underground and 7 elevated stations
(Hiramatsu & Inoue, 2022)

Study Area

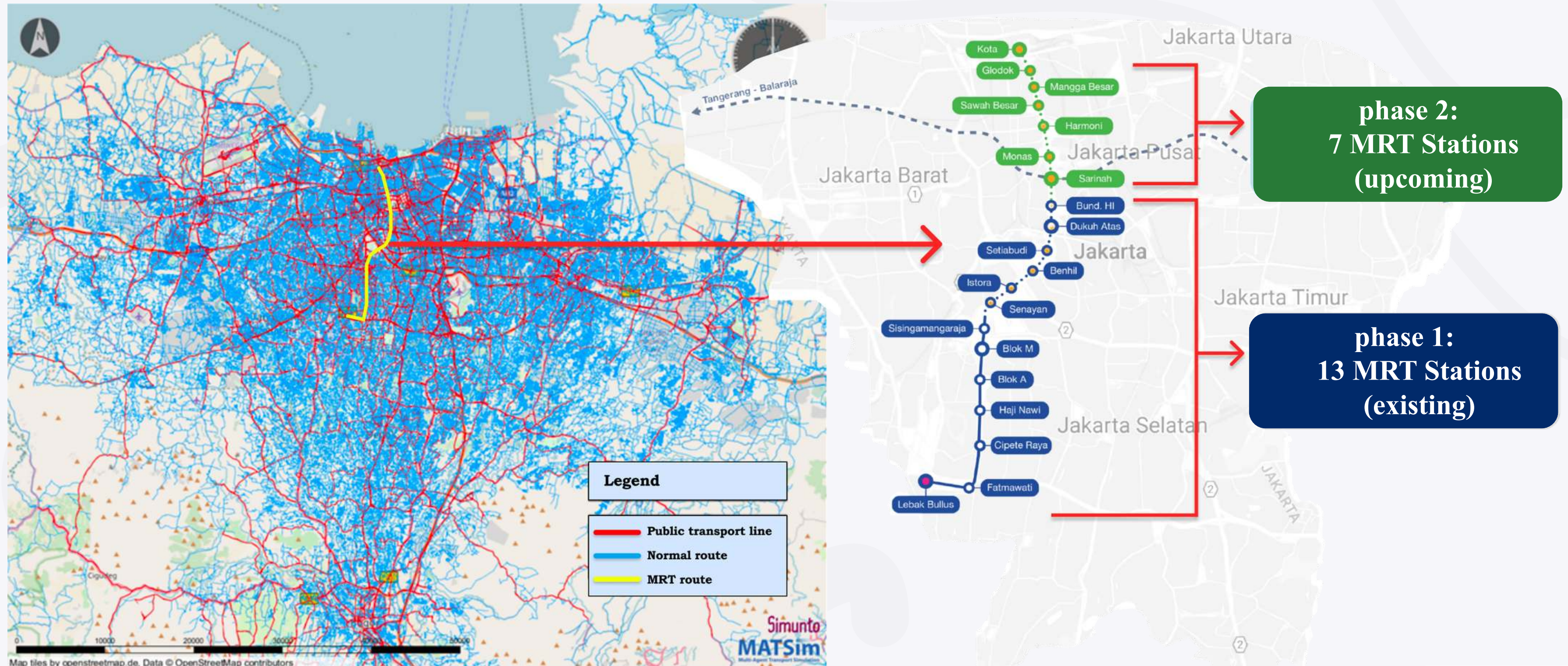


Figure 1. Study Area

Data & Framework

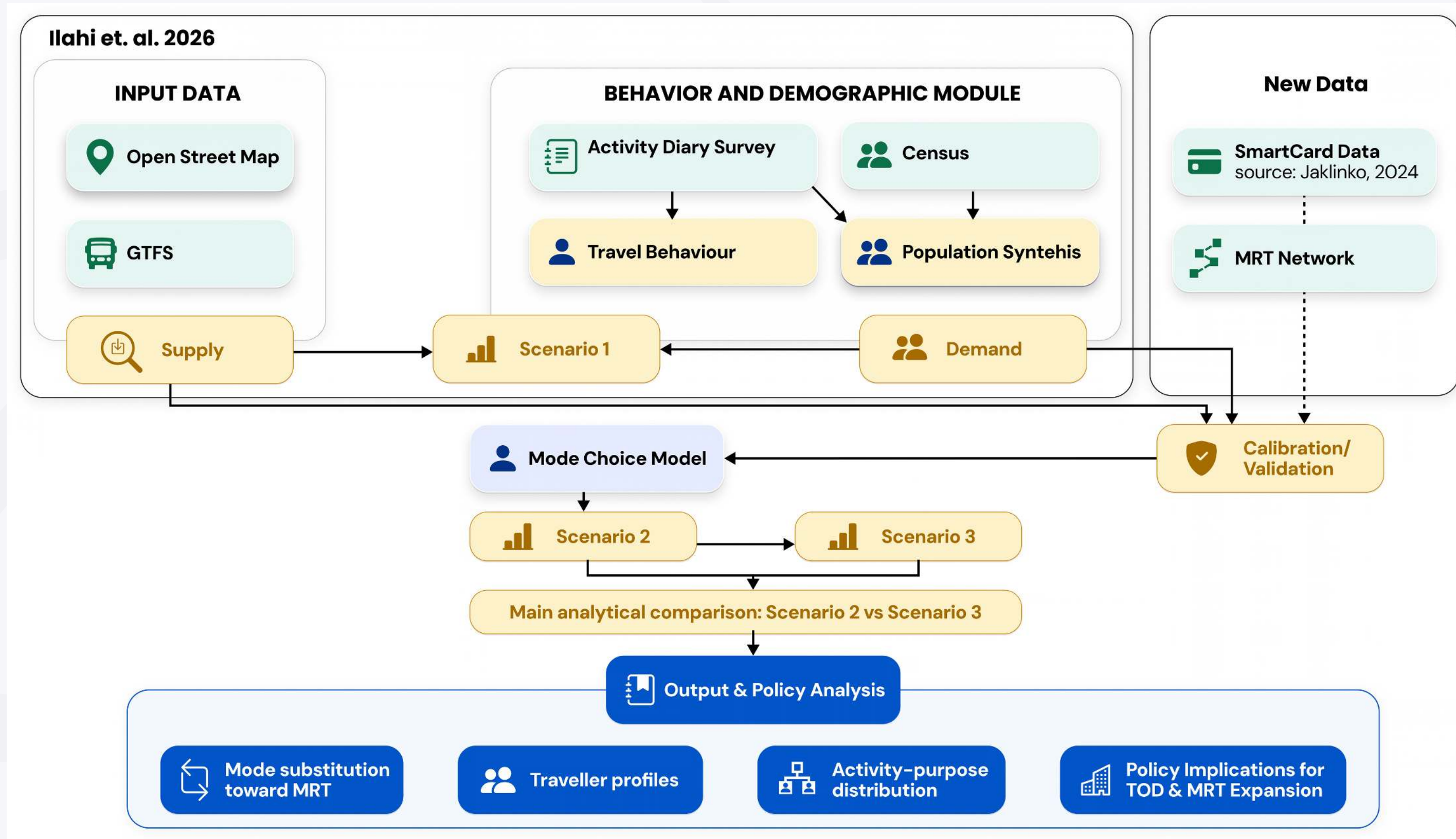


Figure 2. Data & Framework



Methodology

DISCRETE CHOICE MODEL

Behavior mode choice utility specification and calibrated parameter

Behavior Mode Choice Utility Function



Public
Transport
(PT)

$$U_{PT} = ASC_{PT} + \beta_{TT,PT} TT_{PT} + \beta_{Cost} \left(\frac{Distance}{AverageDistance} \right)^{\lambda_{Distance}} \left(\frac{Income}{AverageIncome} \right)^{\lambda_{Income}} Cost_{PT} + \beta_{Wait,PT} Wait_{PT} + \beta_{Access,PT} Access_{PT}$$



Car

$$U_{Car} = ASC_{Car} + \beta_{TT,Car} TT_{Car} + \beta_{Cost} \left(\frac{Distance}{AverageDistance} \right)^{\lambda_{Distance}} \left(\frac{Income}{AverageIncome} \right)^{\lambda_{Income}} Cost_{Car}$$



Motorcycle

$$U_{MC} = ASC_{MC} + \beta_{TT,MC} TT_{MC} + \beta_{Cost} \left(\frac{Distance}{AverageDistance} \right)^{\lambda_{Distance}} \left(\frac{Income}{AverageIncome} \right)^{\lambda_{Income}} Cost_{MC}$$



Walking

$$U_{Walk} = ASC_{Walk} + \beta_{TT,Walk} TT_{Walk}$$



MRT
(This Study)

$$U_{MRT} = ASC_{MRT} + \beta_{TT,MRT} TT_{MRT} + \beta_{Cost} \left(\frac{Distance}{AverageDistance} \right)^{\lambda_{Distance}} \left(\frac{Income}{AverageIncome} \right)^{\lambda_{Income}} Cost_{MRT} + \beta_{Wait,MRT} Wait_{MRT} + \beta_{Access,MRT} Access_{MRT}$$

Where: **ASC** : Alternative-specific constant **Wait** : Waiting time
TT : Travel time **Access** : Access-egress time
Cost : Travel cost **λ** : Distance and income elasticity parameters

Table 7.1. Calibrated Parameters of DMC

Parameters	Estimated value	Calibration
ASC Walking	-7,00	-4,50
ASC PT	-6,00	-3,60
ASC Car	-0,75	-1,00
ASC Car ODT	-6,50	-5,00
ASC MC ODT	-4,00	-2,50
β Travel cost	-2,08	-2,08
λ Income, cost	-0,06	-0,09
λ Distance, cost	-0,75	-0,75
β Travel time Walk	-0,70	-0,70
β Travel time PT	-2,00	-2,00
β Travel time Car	-4,00	-4,00
β Travel time Motorcycle	-6,10	-6,10
β Travel time Car ODT	-8,26	-8,26
β Travel time MC ODT	-7,00	-7,00
β Waiting time PT	-0,05	-0,05
β Access egress time PT	-0,05	-0,05
ASC MRT	-	-4,30
β Travel time MRT	-	-6,00
β Waiting time MRT	-	-0,05
β Access egress time	-	-0,20

(Ilahi, 2026)



Result

Calibration Results



Table 7.2. Aggregate MRT ridership calibration results



Indicator	Value
Observed MRT ridership, January 2024	111,395 trips/day
Equivalent observed ridership at 10% scale	11,139.5 trips/day
Simulated Scenario 2 ridership at 10% scale	12,049 trips/day
Simulated ridership scaled to full population	120,490 trips/day
Absolute difference at full-population scale	+9,095 trips/day
Relative difference	8,16%



Table 7.3. Comparison of simulated daily MRT ridership between the existing 13-station network and the expanded 20-station network.



Mode	Trip		Change
	Scenario 2	Scenario 3	
MRT	12049	15284	3235
Motorcycle	2944	434	-2510
Car	580	95	-485
PT	317	30	-287

Result

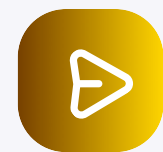
Table 7.4. Temporal distribution of MRT Commuters

Time period	Scenario 2	Scenario 3	Change
Morning, 04:00–09:00	5552	6969	1417
Midday, 09:00–14:00	1458	1776	318
Afternoon, 14:00–19:00	4437	5643	1206
Evening, 19:00–24:00	519	669	150
Time-classified trips	11966	15057	3091

The spatial distribution also broadened following the expansion. Under Scenario 2, MRT Commuters were concentrated primarily around the existing north–south corridor and urban villages in central and southern Jakarta. Scenario 3 produced higher trip concentrations around the planned northern extension and expanded the simulated MRT catchment towards additional urban villages in northern Jakarta. Nevertheless, the strongest demand remained concentrated around the existing corridor, indicating that the new stations complemented rather than replaced the spatial role of the original network.



Figure 3. Matsim Visualisation via Simunto



Result

Temporal distribution of MRT Commuters

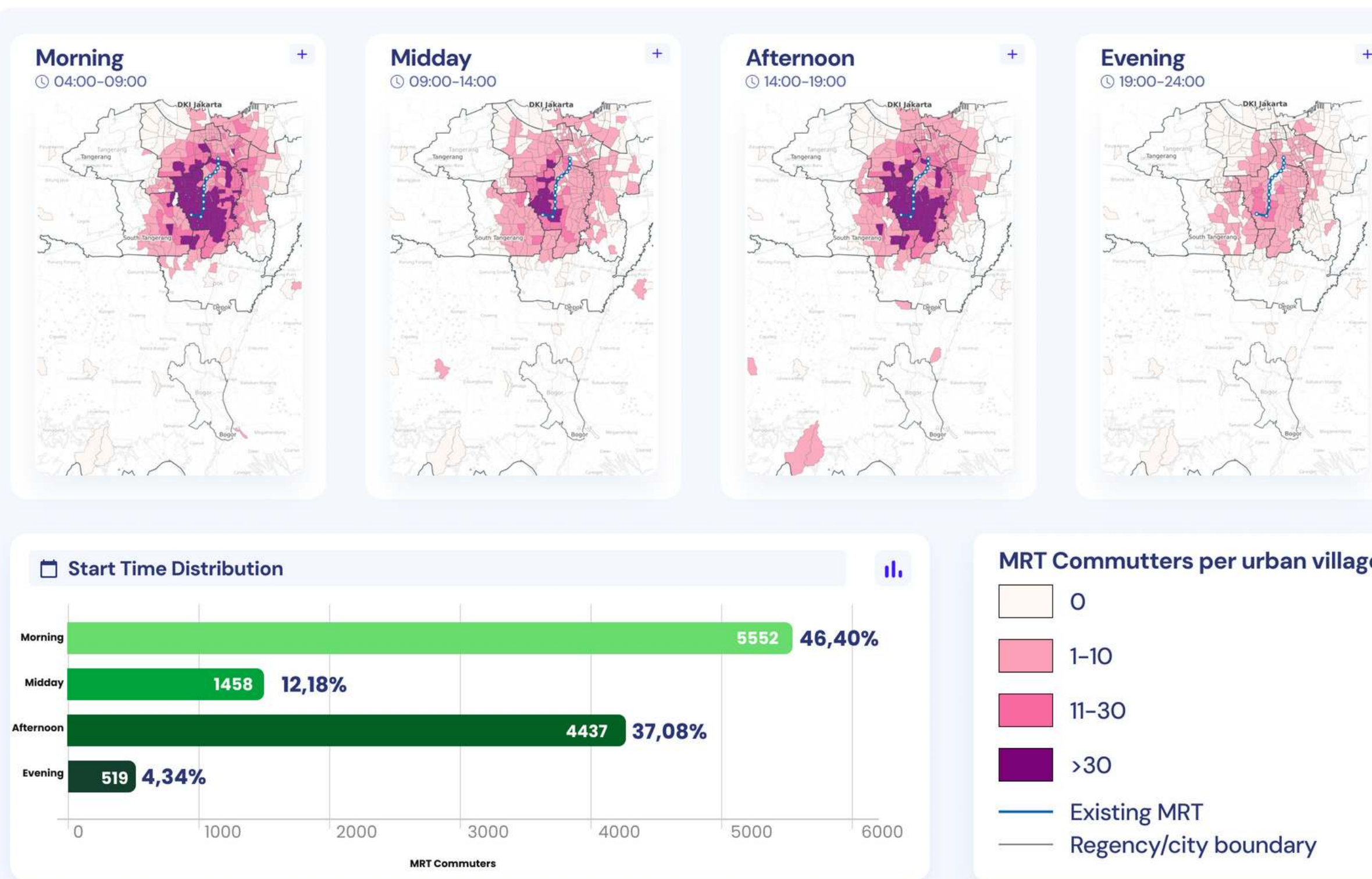


Figure 4. Spatial and temporal distribution of simulated MRT Commuters by urban village under Scenario 2.



Result

Temporal distribution of MRT Commuters

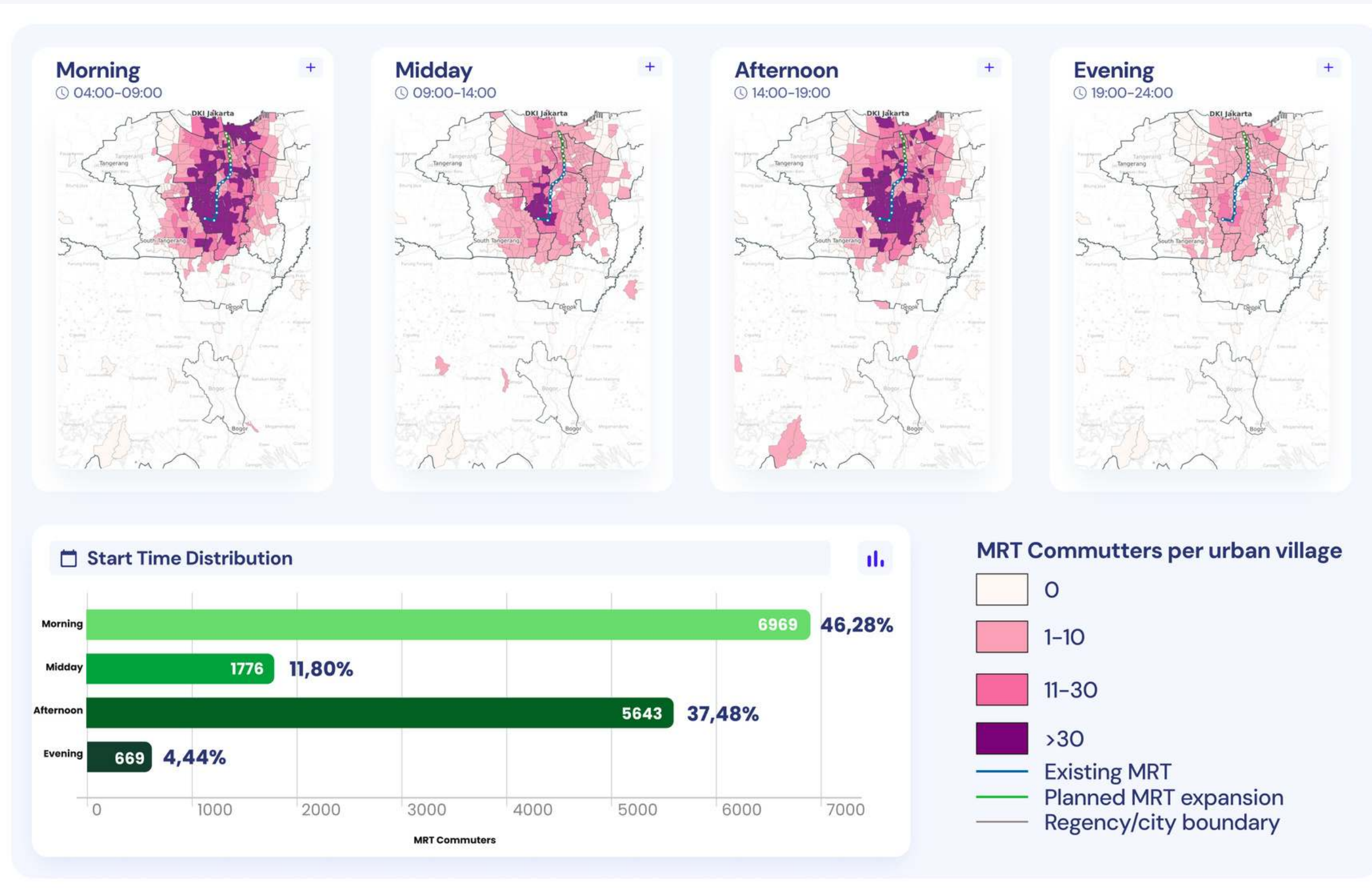


Figure 5. Spatial and temporal distribution of simulated MRT Commuters by urban village under Scenario 3.

Result

Table 7.4. Source modes of trips shifting to MRT between Scenarios 2 and 3.

Mode	Trips	Share (%)	Min (km)	Median (km)	Mean (km)	Max (km)	Std. Dev. (km)
Motorcycle	2,944	7.665	572	885	942	2.317	213
Car	580	151	598	934	976	1.712	214
Public transport (PT)	317	825	589	969	1.016	2.226	256

Table 7.5. Commuter profiles of MRT

Characteristic	Category	Scenario 2	Scenario 3
Age Group (years)	< 18	467	446
	18 - 25	1,17	1,497
	26 - 35	3,298	4,218
	36 - 45	2,96	3,724
	46 - 55	2,186	2,775
	56 - 65	407	514
	> 65	31	43
Gender	Male	6,136	7,717
	Female	4,383	5,5
Monthly Income (thousand IDR)	500	1,071	1,339
	2000	7,093	8,897
	4000	1,793	2,268
	6500	314	406
	10000	158	201
	13500	22	25
	16500	22	26
	19500	4	5
	22000	12	16
	24000	1	1
	30000	29	33

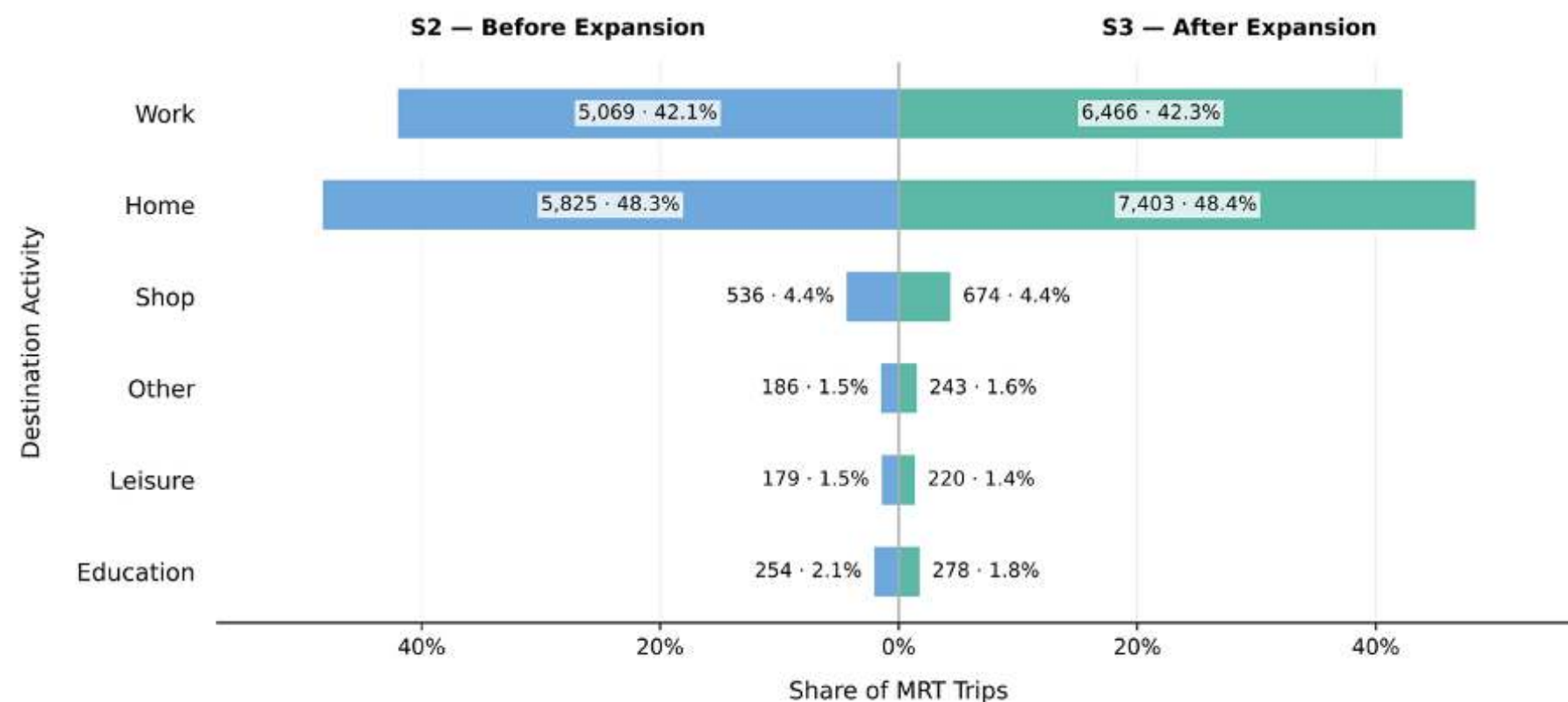


Figure 6. Destination activities of trips shifting to MRT

Finding

- **Ridership increased** by 26.85%, From 12,049 to 15,284 daily trips.
- **MRT coverage expanded** : Accessibility improved, particularly toward northern Jakarta.
- **Most shifts came from motorcycles** : 3,841 shifted trips, 76.65% came from motorcycles, 15.10% from cars, and 8.25% from conventional public transport.
- MRT remained **commute-oriented**: More than 90% of trips were linked to home and work.
- Growth concentrated among **working-age users**: The largest increase occurred among users aged 26–45 years.
- **Expansion increased scale, not user composition**: MRT served more users across a wider area, while the overall user profile remained stable.